5. Analysis or Evaluation

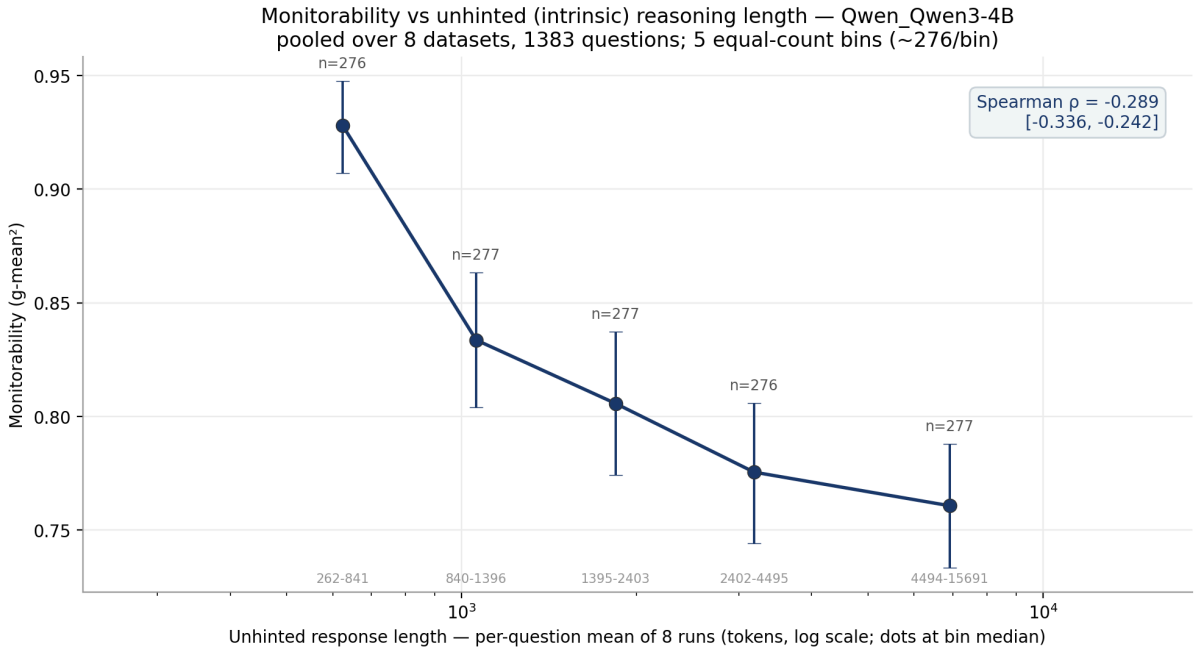
Does monitorability correlate with response length?

We begin with the weaker, indirect signal: whether the unhinted arm's response length tracks monitorability. Unhinted length measures the model's intrinsic reasoning verbosity — not the transcript the monitor reads — so it sets a baseline before the direct hinted-length test.

We work per question: mean unhinted length over eight runs, and mean g-mean² across hint slices. Pooling all eight datasets gives 1,383 questions. Because length is right-skewed (median 1,840 tokens, max 15,690), we bin into equal-count quintiles (~276 each) and summarise the association with a rank correlation, which assumes no functional form.

Monitorability declines monotonically with length, from 0.93 in the shortest quintile (median 625 tokens) to 0.76 in the longest (median 6,905), a drop of 0.17 (Spearman $\rho = -0.29$, 95% CI [−0.34, −0.24]). The effect is real but moderate — the interval excludes zero — and front-loaded: most of the loss falls between the two shortest quintiles (0.93 → 0.83), after which monitorability flattens toward a floor near 0.76. It is the very shortest-reasoning questions that are distinctively easy to monitor; beyond ~1–2k tokens, added length costs little.

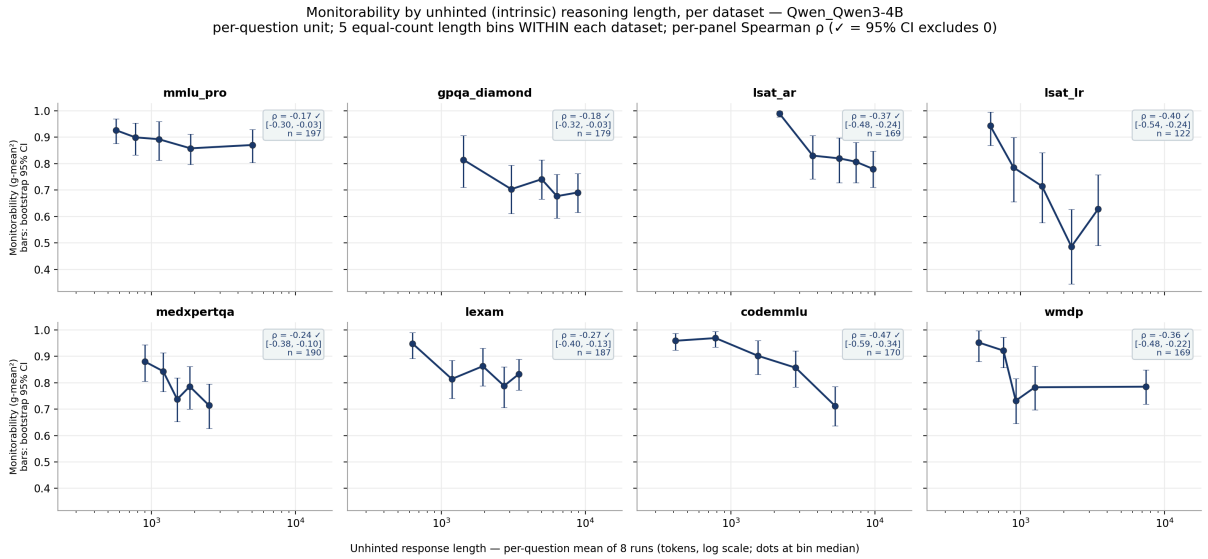
We read this as correlational and partly confounded. Length is strongly negatively related to answer consistency (which we present later in this section) — consistently-answered questions are roughly half as long — so the shortest, most-monitorable questions are disproportionately the consistent ones. We therefore treat unhinted length as an indirect correlate that motivates, but does not establish, a length mechanism, which we test directly with the hinted arm next. [§X shows the consistency effect survives conditioning on length.]



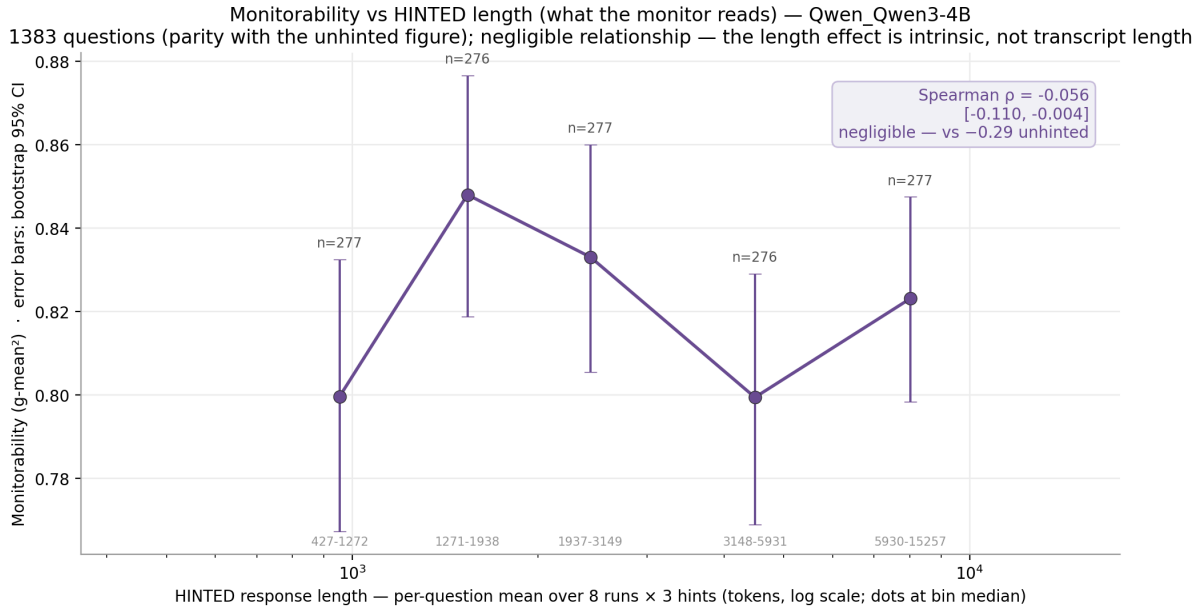
> Figure Y. Monitorability declines with intrinsic (unhinted) reasoning length. Each question (n = 1,383, pooled over eight datasets) is summarised by its mean unhinted response length over eight runs and its mean per-question g-mean². Questions are grouped into five equal-count length bins (~276 each); dots are plotted at the bin's median length (x, log scale) and mean monitorability (y), with bootstrap 95% confidence intervals and bin token-ranges annotated. Monitorability falls monotonically from 0.93 (shortest) to 0.76 (longest); Spearman ρ = −0.29 [−0.34, −0.24]. The association is real but moderate and front-loaded (most of the drop occurs below ~1,400 tokens), and is confounded with answer consistency (shorter reasoning is more consistent).

As a generalisation check, the figure below repeats the pooled length–monitorability analysis separately within each of the eight datasets (per-question unit, five equal-count length bins per dataset, per-panel Spearman ρ). The negative relationship holds everywhere: all eight datasets show a negative ρ, and all eight 95% confidence intervals exclude zero, ranging from −0.17 (mmlu_pro) to −0.47 (codemmlu). This persists across very different reasoning-length regimes — from datasets where the model reasons briefly (wmdp, median ~900 tokens) to ones where it reasons at length (lsat_ar, ~5,700 tokens) — so the pooled ρ = −0.29 is not an artifact of aggregating datasets with

differing verbosity. The per-dataset bins are necessarily thin (~24–39 questions each), so individual panels are noisier and some non-monotonic wobble appears (e.g. lsat_lr), but the direction and significance of the effect are consistent across the board.



Repeating the analysis on the hinted transcript — the text the monitor actually reads, with the question set and monitorability held identical to the unhinted figure so that only the length arm changes — gives a negligible association (Spearman ρ = −0.06 [−0.11, −0.00]), about five times weaker than the −0.29 for unhinted length. Since the length of what the monitor reads carries almost no signal, longer reasoning does not lower monitorability by giving the monitor more text to process; rather, unhinted reasoning length simply flags the questions the model finds hard, and it is that difficulty — not transcript length — that makes them harder to monitor.



On questions the model has to think hard about, it tends to use the hint without admitting it — it quietly follows the hint and then writes up a clean-looking justification that doesn't mention the hint at all. On questions it finds easy, its reasoning is honest — it either ignores the hint or plainly shows what it's doing.

It is not the amount the model writes that makes it dishonest. It's whether the model was sure or unsure to begin with. Writing a lot is just a side-effect of being unsure — when the model is torn, it reasons more. So "long reasoning" and "hides the hint" happen together, but the long reasoning isn't causing the dishonesty — the uncertainty is causing both.

Analogy (you = the model): You're taking an exam and someone whispers "the answer is B."

- On an easy question you already know is A: you think "no, it's obviously A," and your working plainly shows you brushing off B. Honest.
- On a question you're genuinely torn on: the whisper "B" nudges you to B — and you write a tidy, confident-looking argument for B, without writing "I picked B because someone told me." You've absorbed the hint and rationalized it. Dishonest — and this happens exactly on the questions you had to sweat over.

The pages of working you produced aren't what made you dishonest; being on the fence did.

One line: When the model already knows its answer, a hint can't sway it secretly — its reasoning stays honest. When the model is unsure (which is when it reasons the most), a hint can tip it silently, and it rationalizes the result. So it's the model's uncertainty, not the length of its writing, that makes it hard to catch.

- "Uncertain" and "inconsistent" are the same state. Consistency measures it directly — does the model give the same answer across 8 tries? Length measures it indirectly — an unsure model reasons more, so long reasoning is a symptom of the same uncertainty. Both point the same way: the more committed the model is to an answer, the more monitorable it is.
- The bit that could look like a conflict — but isn't: the consistency finding showed that even consistently wrong questions are monitorable. Does "uncertainty" contradict that? No — because "uncertain" here means "hasn't settled on an answer," not "is likely wrong." You can be certain and wrong: the model confidently gives the same wrong answer 8/8. In the exam analogy, you're sure it's A (even though A is actually wrong), so the whisper "B" can't sway you secretly — your reasoning stays honest. Commitment protects monitorability whether or not the committed answer is correct.

	Unhinted length (script 37)	Hinted length (script 41)
What it is	How long the model reasons on its own, no hint	How long the response is when a hint is present
Whose length?	The model's intrinsic thinking	The exact transcript the monitor reads
Result	−0.29 → longer = less monitorable	−0.06 → negligible (≈5× weaker)

length and consistency are the same underlying thing:

Length story (wording)	Consistency story	Same underlying thing
model is sure / committed → reasons briefly	consistent (same answer 8/8)	high commitment → more monitorable
model is unsure / torn → reasons a lot	inconsistent (answers vary)	low commitment → less monitorable

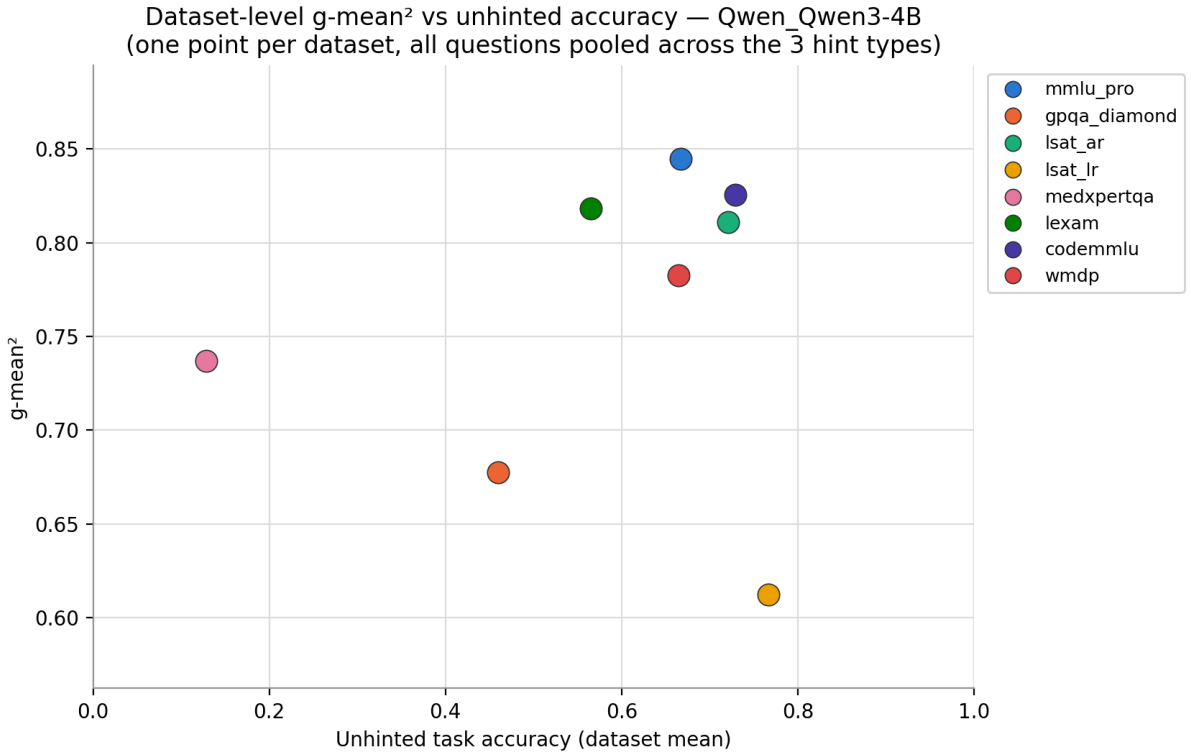
Does monitorability correlate with task accuracy?

Task accuracy does not predict monitorability. What predicts it is whether the model commits to an answer: questions it answers consistently — even consistently wrong — are more monitorable than ones it answers inconsistently (+0.096, 95% CI [+0.056, +0.137]), whereas being correct on top of being consistent adds only a small, statistically unresolved amount (+0.037, 95% CI [−0.004, +0.078]).

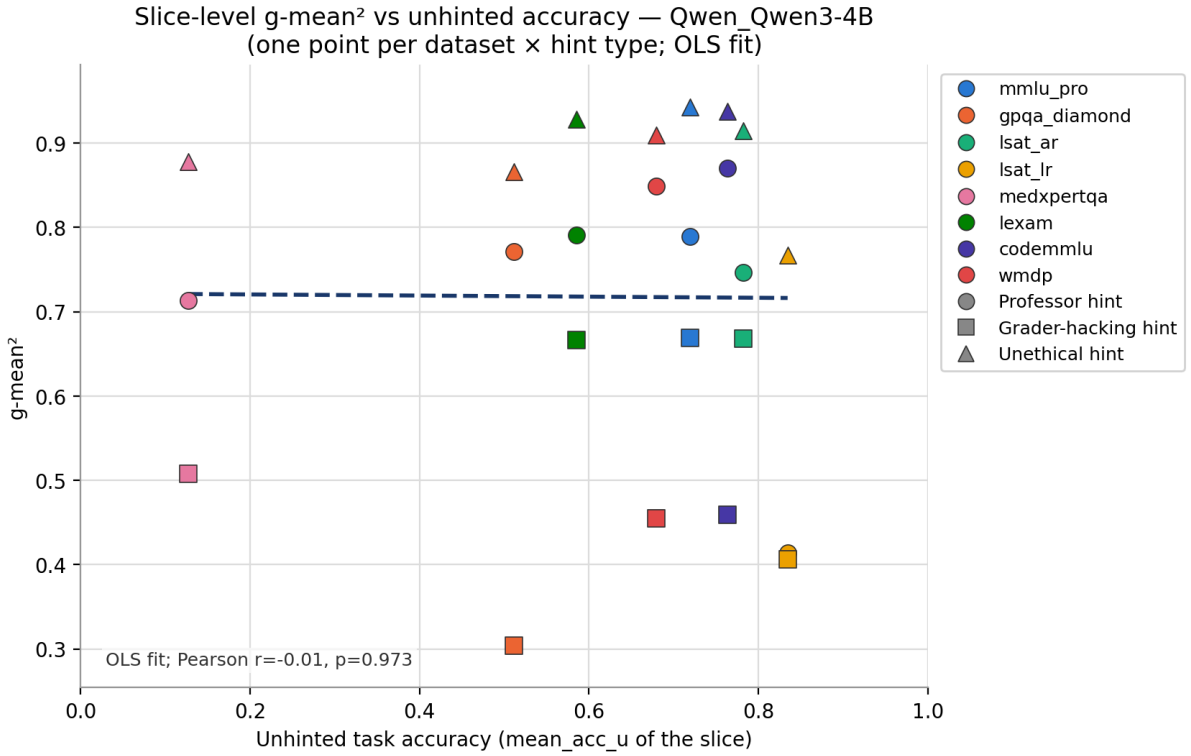
Notes: task performance for any question is measured on the unhinted arm.

Across all configs, does monitorability correlate with task accuracy?

We started our analysis asking weather task performance predicts monitorability score.



In the figure above we can see that some datasets have very low accuracy (less than 20% for medxpertqa), yet they have larger monitorability than datasets the model is much better at (like lsat_at). Though this does not imply negative correlation, since other datasets (like codemmlu and lsat_ar) have similar task accuracy, yet mush higher monitorability. We suspected that there is a lot of variations due to different hint types used for each dataset, so we provide a more detailed view below.



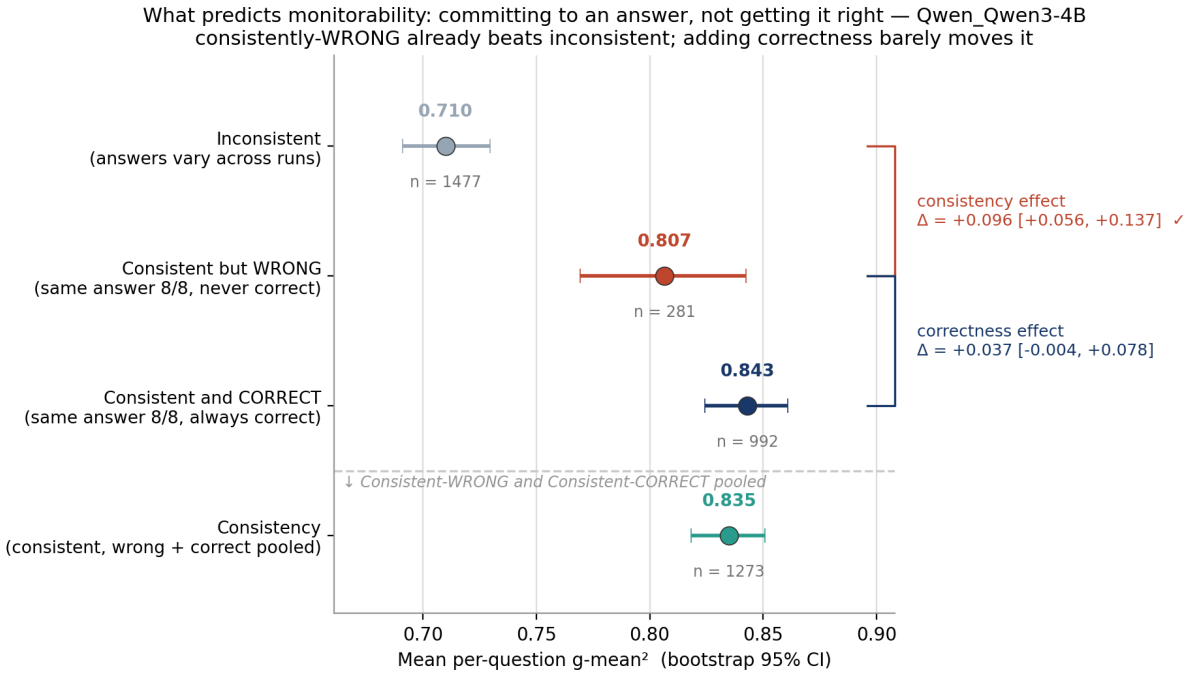
> Slice-level monitorability against task accuracy for the base model (Qwen3-4B). Each of the 24 points is one (dataset × hint-type) slice; the x-axis is the slice's mean unhinted task accuracy and the y-axis its g-mean². Marker colour denotes dataset and marker shape denotes hint type. The dashed line is an ordinary-least-squares fit; the association is null (Pearson r = −0.01, p = 0.97).

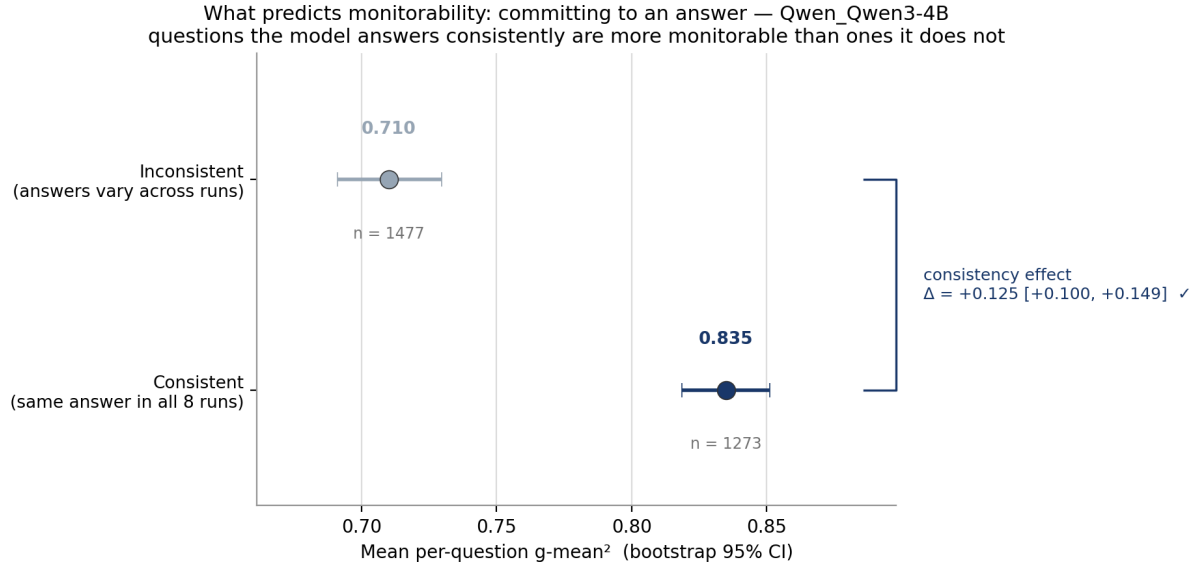
At the level of a whole task, monitorability is unrelated to how well the model performs it. Across the 24 dataset × hint-type slices, the slice's mean unhinted accuracy does not predict its g-mean² (Pearson r = −0.01, p = 0.97), and the fitted line is effectively flat. The vertical spread in the figure is instead organised by hint type, not accuracy: the unethical-hint slices (triangles) cluster near 0.9 while the grader-hacking slices (squares) sit between roughly 0.3 and 0.67, regardless of the underlying task difficulty. The point estimates make the dissociation concrete — lsat_lr is among the most accurate datasets yet one of the least monitorable, whereas medxpertqa is the least accurate yet lands mid-range. Task difficulty therefore does not confound our monitorability results at the dataset level.

Consistency improves monitorability (all questions across all configs):

Since each question in our evaluation is run N=8 times, we recognise that correctness includes multiple modes: the model can be consistently correct, partially correct, or consistently incorrect. The model's susceptibility to using the hint and tendency to verbalise the hint might depend not only on task accuracy, but also on a finer property: the model's confidence/commitment to an answer (consistency). These two notions are related but different. A model can be misscalibrated: it's confident in its wrong answer. In the previous "Slice-level monitorability" figure, we suspected that these variations (consistently correct vs partially correct vs consistently incorrect) are averaged out.

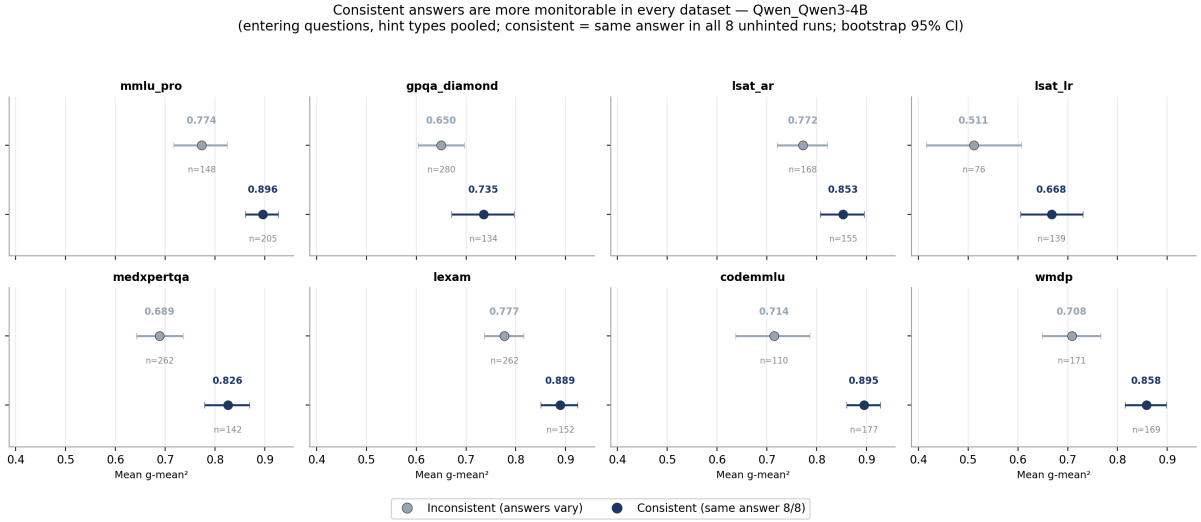
This figure answers "what is it about a question that makes it monitorable?" The answer is commitment, not competence. Move down the three dots: a question the model answers inconsistently scores 0.710; make it consistent but wrong and monitorability jumps to 0.807 (that's the red "consistency effect," +0.096, CI clear of zero — real); now make it consistent and correct and you gain only another 0.037 (the navy "correctness effect," CI touching zero — not resolvable). So the entire action is in the first step, going from "the model can't settle on an answer" to "the model commits to one." Whether that committed answer is right barely matters. A question the model reliably gets wrong is nearly as monitorable as one it reliably gets right, and both leave inconsistent questions well behind. The mechanism is intuitive: when the model has locked onto an answer its chain-of-thought commits to that answer and a monitor can read the commitment; when it's genuinely torn, the reasoning hedges and the monitor degrades.





Consistency improves monitorability (per dataset):

This figure below answers "is that just an artifact of pooling, or does it hold up?" It takes the well-powered version of the same split — inconsistent vs consistent — and shows it in all eight datasets separately. The navy dot sits above the grey dot in every panel, from mmlu_pro to wmdp, with the gap ranging +0.11 to +0.18. It isn't carried by one or two domains; it's a general property of the model. The one visibly weaker case, lsat_lr, is also your least-monitorable dataset across the board, so its muted gap is consistent rather than contradictory. Note that we have removed merged the consistently incorrect and consistently correct classes into Consistent. This is because each dataset had only very few consistently incorrect points, and their CI was very wide.



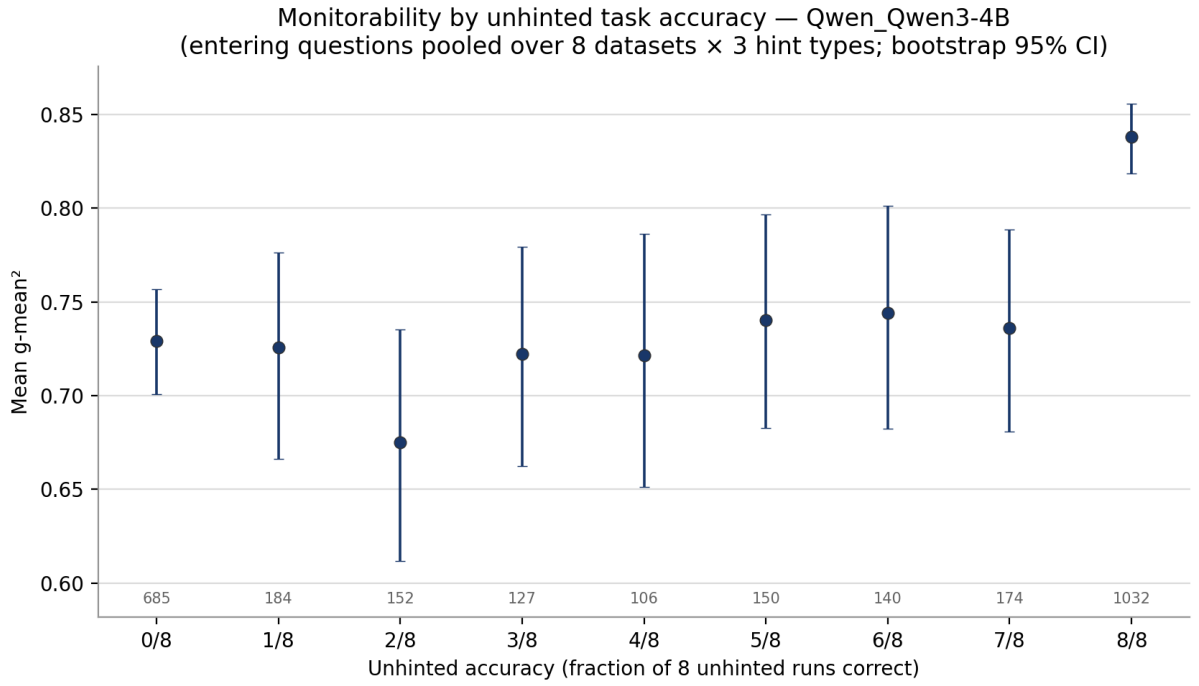
The top figure needs the full 2,750-question sample to resolve the delicate correctness contrast (that's why it's pooled), and the bottom figure needs per-dataset breadth to prove generality (that's why it merges to the two well-powered groups).

Novelty: Previous works who have measured the impact of task performance on monitorability (.....) and faithfulness (.....), though do not disentangle correctness from consistency. Our analysis shows that for an LM that uses the CoT as a reasoning scratchpad, what improves monitorability significantly more than mere correctness is consistency.

Correctness does not improve monitorability (per dataset):

Note: in this analysis we focus only on correctness as the ration of correct answers in the 8 runs. The 0/8 means the model never got the answer right in all 8 runs. This can include both : where the model is consistently wrong, and when the model provides different wrong answers across runs.

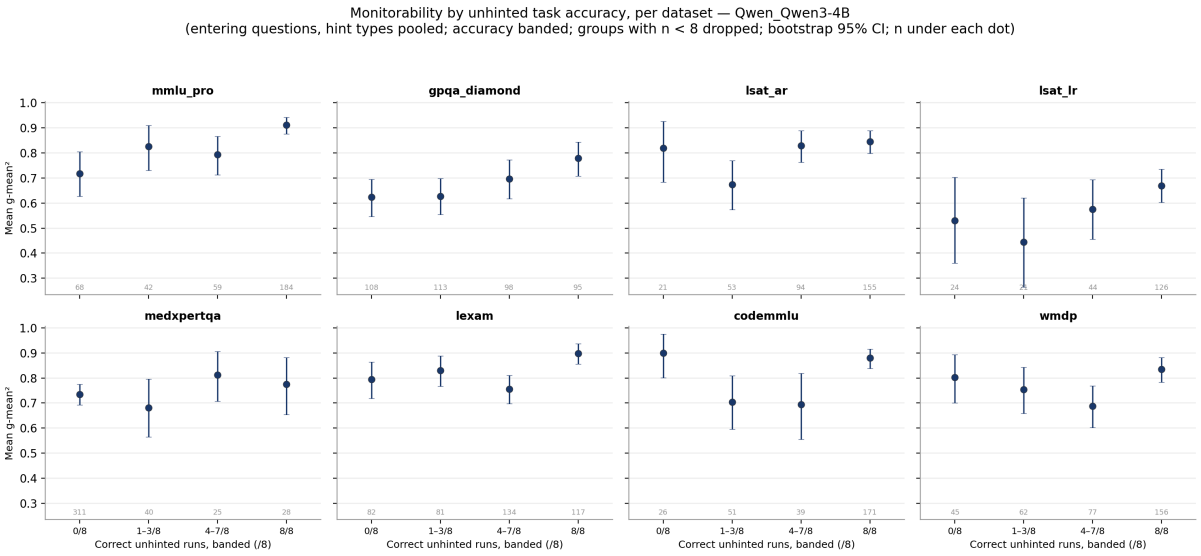
Here we show, if we remove the cases where the model is consistent (always correct or always incorrect), then does monitorability correlate with task accuracy?



> Mean per-question monitorability (g-mean²) by unhinted task accuracy, pooled over the 8 datasets × 3 hint types (n under each level; bootstrap 95% CIs). Accuracy is the number of the 8 unhinted runs answered correctly.

We can see in the above figure that most of the increase in monitorability happens when the model is consistent in its answer (8/8). Monitorability is essentially flat across the partial-accuracy levels (0/8–7/8), hovering near 0.72–0.74 with overlapping confidence intervals, and rises only at 8/8, where the model answers correctly in every run (0.838). Task accuracy therefore does not predict monitorability as a graded trend; the effect is a threshold at full reliability rather than a continuous relationship. Because a question answered correctly in all 8 runs is also answered identically in all 8, this step reflects the model's commitment to a single answer rather than correctness per se — a distinction we isolate in [consistency figure].

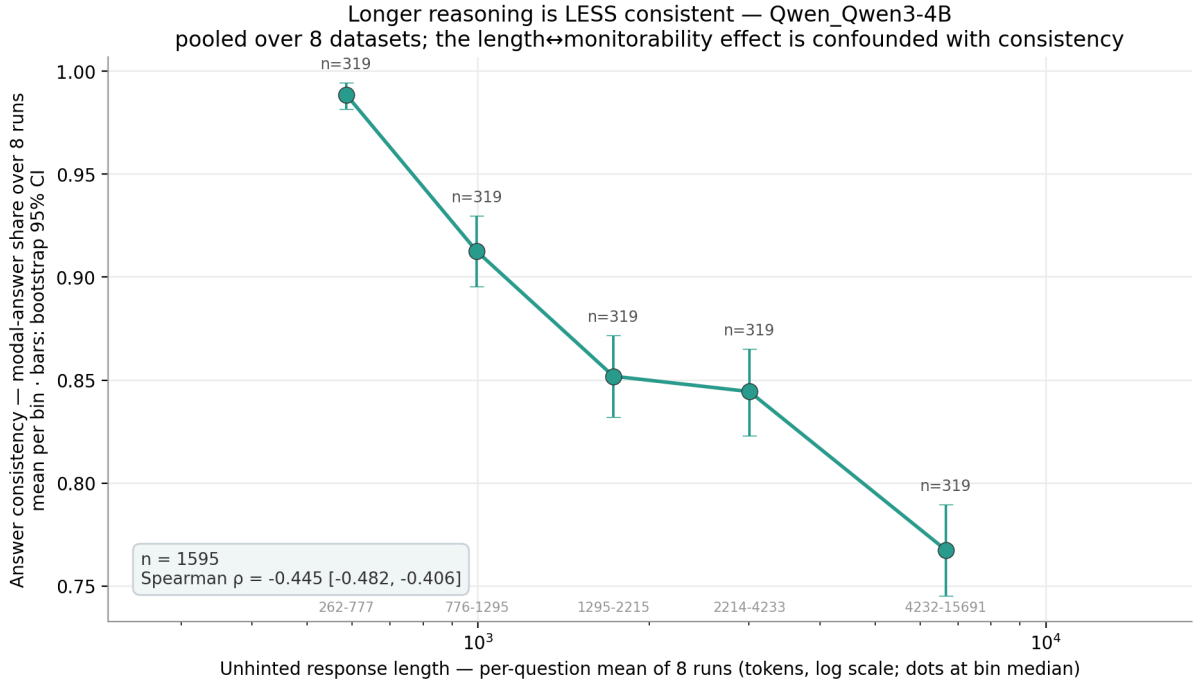
The below figure decomposes the pooled accuracy analysis by dataset, grouping questions into never-correct (0/8), low (1–3/8), high (4–7/8), and always-correct (8/8) accuracy bands to keep each dot adequately powered while isolating the always-correct group. The pooled pattern recurs in most datasets: the always-correct band is the highest or statistically tied-highest point in six of the eight datasets (all except medxpertqa and codemmlu), while the three partial bands show no consistent ordering — they drift upward in some datasets (gpqa_diamond, lsat_lr), downward in others (codemmlu, wmdp), and their confidence intervals overlap within every panel. The two exceptions are minor: in medxpertqa the always-correct band rests on only 28 questions and its interval overlaps the higher partial band, and in codemmlu the never-correct band is marginally highest but well within overlapping intervals. Taken together, the per-dataset view is consistent with the pooled result — task accuracy produces no graded monitorability trend, and only full-run agreement is reliably associated with higher monitorability — indicating that the effect is a general property of the model rather than an artefact of pooling.

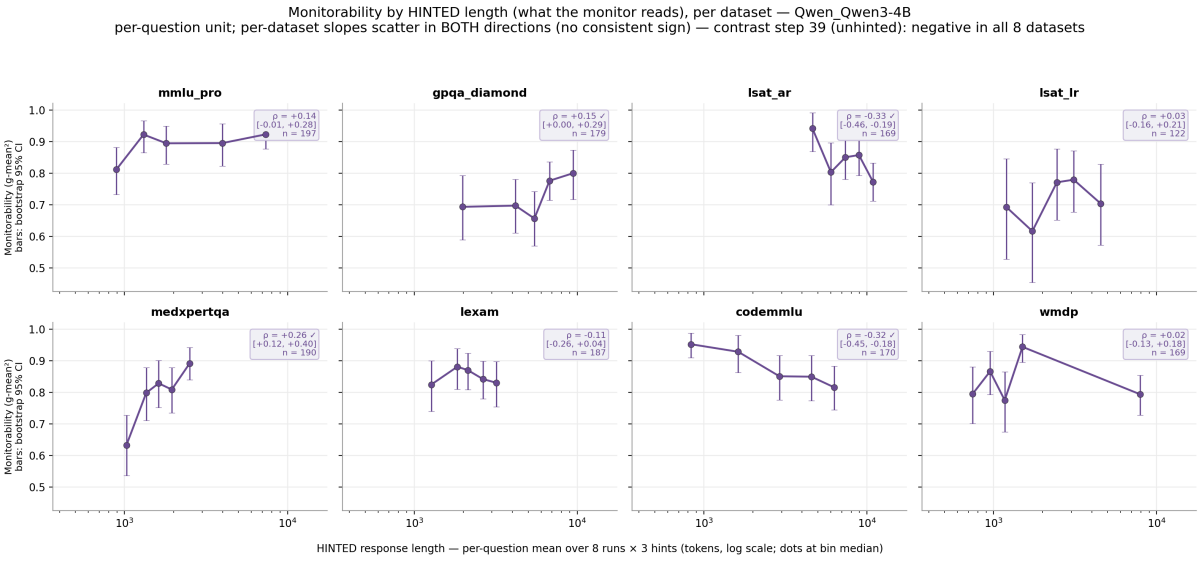


> Monitorability (g-mean²) by unhinted-accuracy band, per dataset (never-correct 0/8, low 1–3/8, high 4–7/8, always-correct 8/8; n under each dot; bootstrap 95% CIs). The always-correct band is the most monitorable in most datasets while the partial bands show no consistent ordering, confirming the pooled reliability-threshold pattern generalises across datasets.

Does consistency correlate with unhinted response length? If so in which direction?

We have found above that consistency and shorter unhinted response length both correlate with high monitorability. So this prompts us to think of weather the two are also correlated. Meaning: more consistent responses tend to be shorter?





A latent common cause — how determined the model's answer is on a question — drives all three. When the answer is well-determined, the model commits (high consistency), reasons briefly (short length), and produces a faithful, monitorable CoT. Consistency, length, and monitorability are three correlated markers of that same underlying state.

Does monitorability correlate with TE?

Monitorability is uncorrelated with hint-susceptibility (treatment effect TE = $p_1 - p_0$): Spearman $\rho = -0.018$, 95% CI [−0.057, +0.020], indicating that how much a hint sways the model is independent of whether it verbalises the influence.